

KUMAMOTO, Japan

WMO: 478190

Lat: 32.8136N

Lon: 130.7069E

Elev: 39

StdP: 100.85

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.6	-0.5	-7.7	2.0	4.5	-6.4	2.2	4.3	7.5	6.2	6.7	6.5	1.7	10	0.372

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.8	34.7	25.5	33.7	25.2	32.7	24.9	26.7	31.6	26.2	30.9	25.9	30.3	3.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.5	20.8	28.8	25.1	20.3	28.5	24.6	19.7	28.2	83.9	31.7	81.9	31.1	80.2	30.4	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.1	5.3	-4.1	36.3	1.2	0.9	-5.0	36.9	-5.7	37.5	-6.4	38.0	-7.3	38.6		
			-4.9	27.5	1.2	0.6	-5.8	27.9	-6.5	28.2	-7.2	28.6	-8.1	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	5.9	7.5	10.9	16.0	20.8	24.1	28.0	29.0	25.6	20.1	13.7	8.1	(6)
(7)		DBStd	8.40	2.74	3.40	3.23	3.11	2.26	2.09	2.08	1.75	2.58	3.00	3.51	3.24	(7)
(8)		HDD10.0	330	129	87	28	1	0	0	0	0	0	7	78	(8)	
(9)		HDD18.3	1483	384	304	232	82	5	0	0	0	16	142	317	(9)	
(10)		CDD10.0	3077	3	16	54	180	336	422	559	589	468	312	119	19	(10)
(11)		CDD18.3	1188	0	0	0	11	82	172	301	331	218	69	4	0	(11)
(12)		CDH23.3	11162	0	0	2	62	482	1128	3272	3922	1917	367	10	0	(12)
(13)		CDH26.7	4196	0	0	0	4	88	289	1325	1764	664	62	0	0	(13)
(14)	Wind	WSAvg	2.2	2.0	2.2	2.3	2.3	2.3	2.5	2.4	2.2	2.1	1.9	1.9	(14)	
(15)	Precipitation	PrecAvg	2255	68	97	151	172	198	479	410	219	194	99	97	72	(15)
(16)		PrecMax	3737	161	194	380	287	417	963	960	555	528	218	211	122	(16)
(17)		PrecMin	1145	12	37	68	78	56	107	71	54	32	7	14	2	(17)
(18)		PrecStd	489	34	42	63	67	88	227	225	128	125	60	50	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	20.1	23.0	27.2	30.6	32.3	35.7	36.1	34.4	30.4	24.8	19.0	(19)
(20)		MCWB	12.7	14.7	14.8	17.5	18.8	22.8	25.7	25.8	24.9	22.6	18.1	13.8	(20)	
(21)		2%	DB	14.1	17.4	20.6	25.1	28.8	30.9	34.4	35.0	32.9	28.4	22.7	16.9	(21)
(22)		MCWB	10.3	13.0	13.4	16.1	18.6	22.1	25.4	25.7	24.2	20.7	16.8	12.5	(22)	
(23)		5%	DB	12.6	15.3	18.6	23.5	27.5	29.6	33.4	34.2	31.7	26.8	21.1	15.3	(23)
(24)		MCWB	8.9	11.4	12.8	15.6	18.2	21.7	25.3	25.4	23.7	19.7	15.9	11.7	(24)	
(25)		10%	DB	11.2	13.6	16.9	21.9	26.2	28.4	32.3	33.2	30.4	25.4	19.6	13.8	(25)
(26)		MCWB	7.8	9.8	12.3	15.3	17.8	21.7	25.2	25.2	23.2	19.1	15.5	10.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	16.4	17.3	21.0	23.1	25.7	27.3	27.4	26.4	24.4	20.2	15.8	(27)
(28)		MDCB	15.7	18.0	19.0	23.3	26.1	29.1	32.7	32.8	30.9	27.4	22.0	16.7	(28)	
(29)		2%	WB	11.6	14.2	15.7	19.3	21.9	25.0	26.7	26.8	25.6	23.0	18.6	13.8	(29)
(30)		MDCB	12.8	16.5	18.1	22.0	24.5	27.9	31.5	31.8	30.0	26.5	20.5	15.5	(30)	
(31)		5%	WB	9.6	12.3	14.3	18.0	20.8	24.5	26.2	26.4	25.0	21.7	17.4	12.2	(31)
(32)		MDCB	11.8	14.7	17.4	20.7	23.7	27.2	30.7	31.3	29.2	25.1	19.9	14.5	(32)	
(33)		10%	WB	8.2	10.3	13.0	16.7	19.8	23.8	25.8	26.0	24.4	20.4	16.2	10.9	(33)
(34)		MDCB	10.8	12.7	16.1	20.0	23.4	26.4	30.0	30.8	28.5	23.7	19.2	13.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.4	9.0	9.5	10.0	9.6	7.3	7.1	7.8	8.1	9.2	9.1	8.7	(35)	
(36)		5% DB	MCDBR	10.4	11.0	11.8	12.4	12.0	9.7	9.0	9.2	9.4	10.6	10.2	10.2	(36)
(37)		MCWBR	7.1	7.2	6.7	5.7	4.7	3.6	2.9	2.9	3.4	4.5	5.7	6.7	(37)	
(38)		5% WB	MCDBR	8.8	9.0	9.1	8.0	7.5	5.7	7.8	8.0	7.6	7.5	7.6	8.2	(38)
(39)		MCWBR	7.2	7.2	6.7	5.1	4.5	2.9	2.9	2.8	3.1	3.9	5.2	6.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.422	0.461	0.521	0.540	0.551	0.559	0.555	0.529	0.500	0.454	0.421	0.413	(40)	
(41)		taud	2.086	2.018	1.868	1.875	1.916	1.976	2.029	2.093	2.136	2.179	2.182	2.128	(41)	
(42)		Ebn at Noon	763	775	759	766	762	752	753	767	771	776	762	748	(42)	
(43)		Edh at Noon	131	155	194	200	194	182	172	159	146	130	118	119	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.37	3.04	4.06	4.80	5.11	4.20	4.89	5.09	4.22	3.57	2.69	2.19	(44)	
(45)		RadStd	0.23	0.27	0.26	0.44	0.57	0.64	0.66	0.48	0.46	0.38	0.27	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page